

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	PE-V : Lab : Blockchain Technology
Semester	7	Course Code	23IOT1748
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	07
Course Outcome	1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain. 2. Outline various research aspects for development of blockchain. 3. Make use of different security protocols to build secure blockchain applications. 4. Analyze smart contracts and Hyperledger Fabric used in blockchain. 5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.
Aim	Develop a Smart Contract for Basic Cryptocurrency Transactions (Sender, Receiver, and Amount).
Theory (100 words)	A Smart Contract is a computer program deployed on a blockchain that automatically executes predefined operations when specified conditions are satisfied. It eliminates the need for a third party by using blockchain technology to record and process transactions in a transparent and secure manner. Smart contracts are commonly developed using Solidity on the Ethereum blockchain. In this experiment, a smart contract is designed to demonstrate a basic Ether (ETH) transfer between two blockchain addresses. The contract maintains transaction information such as the sender address, receiver address, transferred amount, and transaction details. The payable function allows the contract to accept Ether along with a transaction. Before processing the transfer, appropriate conditions are checked to ensure that the receiver address is valid and that the amount being transferred is greater than zero. Once the conditions are satisfied, the specified amount of Ether is transferred to the receiver's address and the transaction information is recorded in the contract. An array can be used to maintain multiple transaction records. Functions such as <code>getTransaction()</code> can retrieve the details of a particular transaction, while <code>getTotalTransactions()</code> can provide the number of transactions stored in the contract. The execution of a smart contract is recorded on the blockchain, making the transaction history transparent, traceable, and difficult to modify. Each



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	transaction also requires gas , which represents the computational resources required to execute the contract. Thus, this experiment demonstrates the basic concepts of Ether transfer, blockchain addresses, payable functions, transaction recording, validation, and smart contract execution.
Procedure and Execution (100 Words)	1. Open Remix IDE and create a new Solidity file named Practical7_iot61.sol. 2. Write or paste the BasicTransaction smart contract code into the Solidity editor. 3. Select Solidity compiler version 0.8.20 or another compatible 0.8.x version and compile the contract. 4. After successful compilation, open Deploy & Run Transactions from the left sidebar. 5. Select Remix VM as the execution environment and choose the BasicTransaction contract. 6. Click the Deploy button to deploy the smart contract. 7. After successful deployment, locate the contract under Deployed Contracts. 8. Select a Remix VM account to act as the sender. 9. Enter another valid Remix VM account address as the receiver in the sendMoney() function. 10. Enter the required Ether amount in the Value field and select the appropriate unit, such as wei or ether. 11. Click sendMoney() to execute the transaction and transfer the specified amount. 12. Verify that the transaction information, including the sender address, receiver address, and transferred amount, is stored in the transactions array. 13. Call getTotalTransactions() to verify the total number of transactions recorded by the smart contract. 14. Use getTransaction(0), getTransaction(1), and so on to retrieve and verify individual transaction details. 15. Finally, check the Remix terminal/logs to confirm successful transaction execution and observe the associated gas details.



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Stepwise Screenshots:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract practical7_iot61 {

    struct Transaction {
        address sender;
        address receiver;
        uint amount;
    }

    Transaction[] public transactions;

    function sendMoney(address payable receiver) public payable {
        require(receiver != address(0), "Invalid address");
        require(msg.value > 0, "Amount must be greater than 0");
        receiver.transfer(msg.value);
        transactions.push(
            Transaction(msg.sender, receiver, msg.value)
        );
    }
}
```

DEPLOY & RUN TRANSACTIONS

Environment: Remix VM, Osaka

Account 1: 99.999 ETH

Deploy: practical7_iot61 (Compiled)

Value: 0 wei, Gas Limit: auto

Deploy



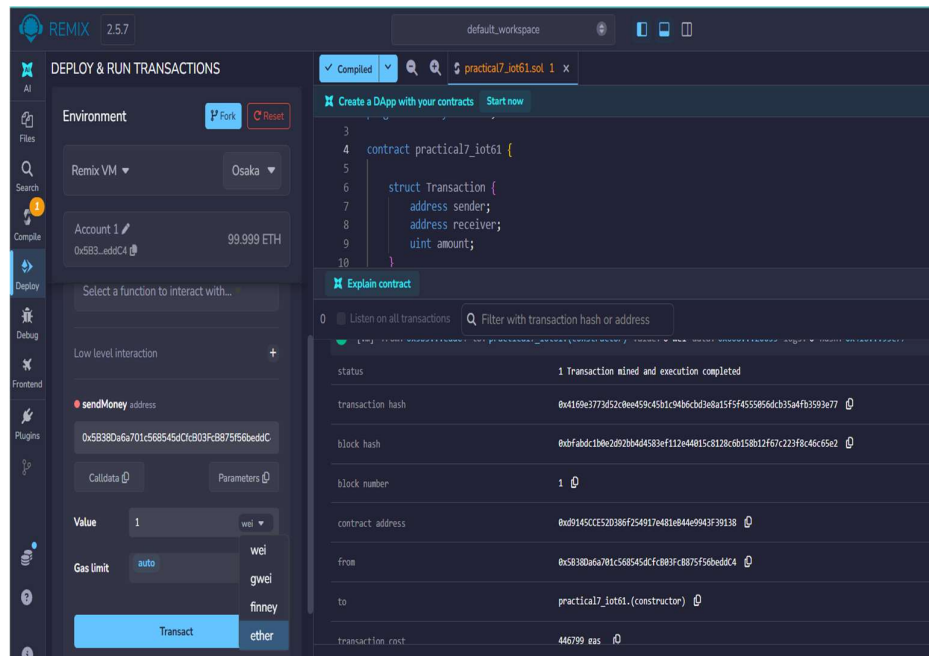
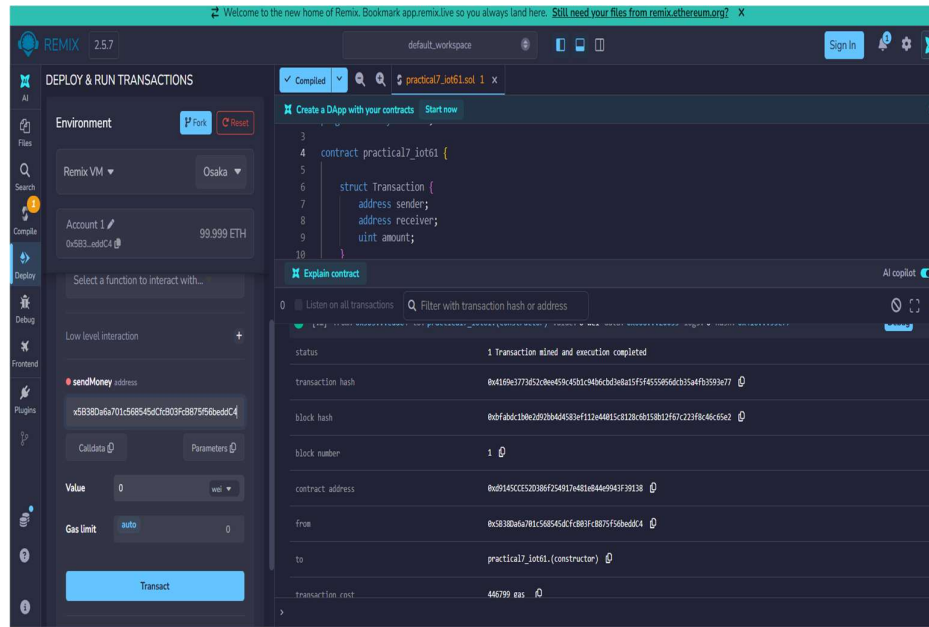
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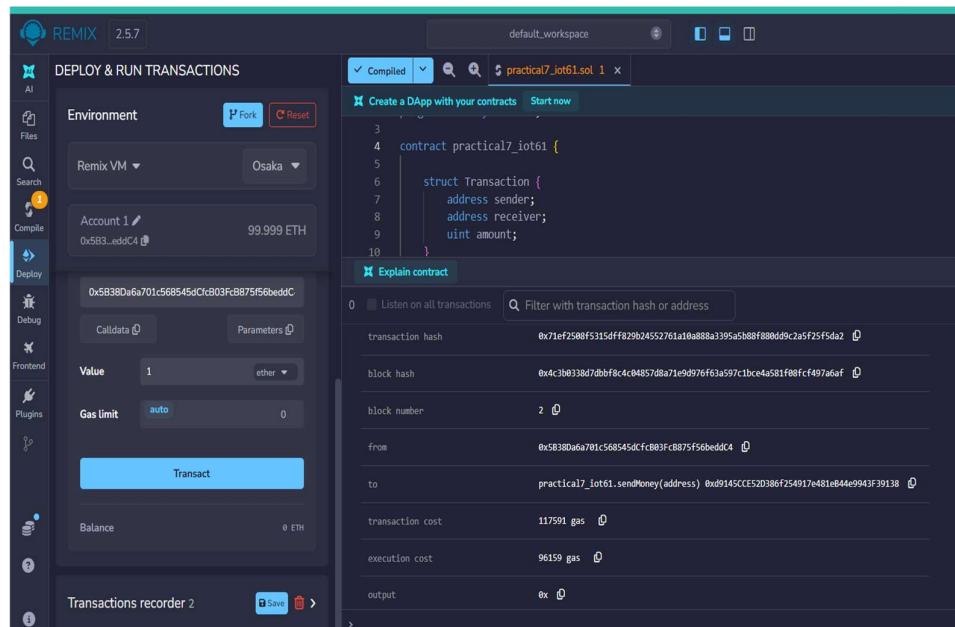
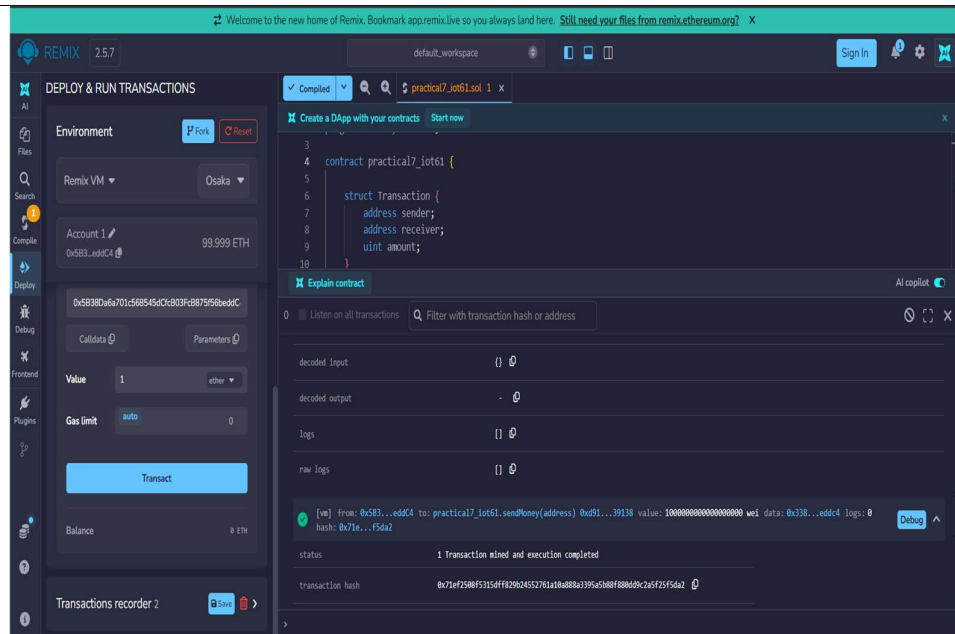
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Code	<pre>// SPDX-License-Identifier: MIT pragma solidity ^0.8.20; contract practical7_iot61 { struct Transaction { address sender; address receiver; uint amount; } Transaction[] public transactions; function sendMoney(address payable receiver) public payable { require(receiver != address(0), "Invalid address"); require(msg.value > 0, "Amount must be greater than 0"); receiver.transfer(msg.value); transactions.push(Transaction(msg.sender, receiver, msg.value)); } function getTransaction(uint index) public view returns (address, address, uint) { Transaction memory t = transactions[index]; return (t.sender, t.receiver, t.amount); } function getTotalTransactions() public view returns (uint) { return transactions.length; } }</pre>
Conclusion	The Basic Cryptocurrency Transaction smart contract was successfully developed and executed using Solidity in Remix IDE. The contract demonstrated the transfer of Ether between two valid blockchain addresses while maintaining a record of each transaction. The sender address, receiver address, and transferred amount were successfully stored and retrieved using the contract functions. The practical also demonstrated the use of payable



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	functions, address validation, transaction storage, and gas consumption. Hence, the experiment provided a clear understanding of how smart contracts can automate and record cryptocurrency transactions securely on a blockchain.
Date	08/09/2026